

Surface modifications for cryogenically conditioned copper electrodes

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Abstract

Different sets of copper electrodes were conditioned at different temperatures. The electrodes showed atypical morphological features. The electrodes were tested at cryogenic temperatures. In this report, the results of the tests at different temperatures are presented.

rewrite:

the ability to sustain electric fields limited. It was previously shown that low-T conditioning leads to improved conditioning process. here we analyse features formed during conditioning and compare features observed. It is shown that although the limiting properties of the samples are similar, there are significant differences in the forms of the BD

1. Introduction

2. Methodology

2.1. Copper samples

The material for all the electrodes compared in this paper was the OFHC copper (C10100 Grade 1), the same material used for production of CLIC cavities [1]. The electrodes were produced using fly-cut with further diamond turning, achieving surface roughness of less than 25 nm. All electrodes followed the cleaning procedures with the chemical solvents, as the accelerating structures for CLIC project [2]. In the results presented below, we refer to the *hard* copper and *soft* copper. The "hard" copper samples came directly from the mechanical production and are cleaned chemically. The "soft" copper samples follow the same production cycle as hard copper, but are additionally annealed. They differ visually, as typically one can find regular, microscopic, grooves after diamond machining on hard copper samples with small grain sizes (20-80 μm). The grooves are gone from the soft copper surface due to high annealing temperature (1040°C). In addition, one can distinguish the soft Cu surface by much larger crystal grain regions (few mm).

2.2. Electrodes

Results from four different set of electrodes are discussed in this paper. Details of the copper samples and location of the tests are shown in table 1. All the electrodes pairs have the same geometry. During the experiments, the electrodes are separated by a ceramic spacer (AlO_3 , alumina) with a high dielectric strength. The spacer and the electrodes are machined very precisely, such that the distance between the electrodes is kept constant at the level of tens of μm . The cathode surface has 62 mm diameter and the anode has 40 mm diameter.

Table 1: Details of the copper samples and test location.

Electrodes	Copper type	Test location	Sample preparation process	Temperature	Gap, μm
015 Cu	soft Cu	CERN	Heated up to 800°C in hydrogen atmosphere and baked up to 650°C	room temperature	100
					60
					40
					20
035 Cu	soft Cu	Uppsala	Heated up to 1040°C in hydrogen atmosphere and baked up to 650°C	room	40
				30 K	59
038 Cu	hard Cu	Uppsala, Jerusalem	None	30 K	59
052 Cu	hard Cu	Uppsala	None	10 K	59

2.3. Conditioning

The copper sample 015 was conditioned at the CERN dc discharge system, while the copper samples 035, 038 and 052 were conditioned at the cryogenic dc discharge system in Uppsala. The standard conditioning procedure developed for the rf and dc tests for CLIC project [3–5] was followed in all the cases. In the cold dc system, the temperature of the electrode can be controlled, and the same electrodes are routinely conditioned at different temperatures. The details of the cryogenic system can be found in [6].

During the conditioning, $1\ \mu\text{s}$ high-voltage pulses and 1 kHz repetition rate are used. The conditioning test is done using feedback mode, i.e. the pulsing started with a low voltage (few hundred volts), and increased by a set increment of $dV = 9\text{V}$ if no breakdown occurred over $n_{\text{max}} = 100000$ pulses. This continues until a breakdown is detected. At BD, the voltage drops to $1/5$ of its last value and is asymptotically increased up to V_{BD} in 20 steps, with 100 pulses at each step. If the breakdown occurs again, the target voltage is decreased until another stable voltage level is found. The details are in [3]. The breakdown rate is monitored during the conditioning and is typically kept at 1×10^{-5} BD/pulse. Breakdown rate is calculated using an exponentially weighted moving average method over a 1 million pulses window. Electrodes 015 were only conditioned at room temperature but the gap distance was varied in different runs to 20, 40, 60, 100 μm . Electrodes 035, with the gap of 40 μm were, first conditioned at room temperature and then cooled down to 30K for experiments at cryogenic temperature. The electrodes 038 and 052, with the same gap, were directly cooled down to 30 K and 10K respectively for conditioning. After full conditioning, no additional experiments were performed on the electrodes. It should be mentioned that the gap between electrodes increases after cooling to

50 cryogenic temperatures from 40 to 59 μm due to different expansion coefficients of the ceramic and
51 copper.

52 2.4. Analysis

53 2.4.1. Surface microscopic observation in Uppsala

54 The microscopic observations of the breakdown features on anode and cathode were done using
55 SmartScope FLASH 200 light microscope available in Uppsala University. First, the corresponding
56 areas on anode and cathode were scanned with 1 $\mu\text{m}/\text{px}$ resolution. The single scan were then stitched
57 together to form a larger map. Next, the analysis of the larger maps was done using a dedicated
58 MATLAB script, similar to one described in [4]. An automatic MATLAB circle-finding routine was
59 used at the initial stage to create an array with the coordinates of the recognized surface features with
60 first estimate for the circles' radii. This was prepared separately for anode and cathode for further
61 analysis. A link between the anode and cathode images and coordinates was done semi-manually
62 and resulted in a set of the transformation factors (a scale, a rotation angle and translation x and
63 y positions) that allowed for direct comparison of the features on anode and cathode for the same
64 breakdown. The software fits simultaneously two circles on top of the BD feature for the anode (the
65 red circles on the left part of fig. 1) and the cathode (blue circles on the right part in fig. 1), allowing
66 however for manual adjustment of the size and coordinates, and links them as an anode-cathode couple
67 in a database. Manual control was very important since some of BD features can sit close to each
68 other and automatic recognition routine can miss it. The irregular features are also accounted for, but
69 the size found automatically by the algorithm had to be always adjusted. These features on cathode
70 and corresponded anode's features are highlighted with yellow circles on fig. 1.

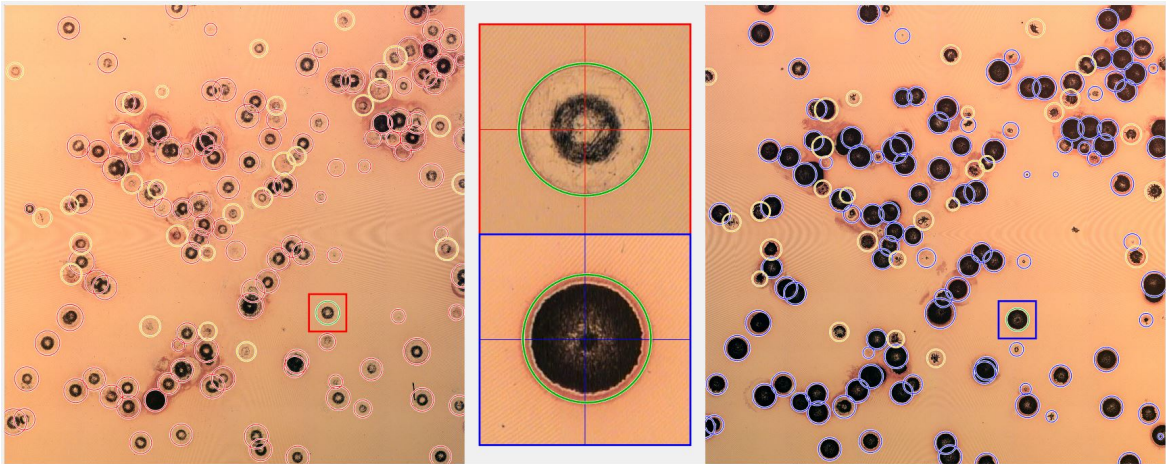


Figure 1: A front panel of program for counting breakdowns on electrodes: anode is on the left, the cathode is on the right. In the middle, is a zoom of selected part on anode and corresponded on cathode. 038 Hard Cu is shown.

71 2.4.2. Surface analysis in Jerusalem

72 The 038 pair of electrodes tested at cold dc system was sent for surface investigations to the Hebrew
73 University of Jerusalem. Characterization with light microscopy, SEM, FIB and TEM was performed.

The hardness of the samples was measured at different locations on the electrode surface, looking for difference between soft and hard copper before and after the conditioning.

2.4.3. Time and voltage relation

The dc system at CERN is equipped with a BD localization system [4], that uses triangulation of the emitted light during BD in the system to determine where the discharges have occurred and match it to the microscopic feature left on the surface. The system can only operate at room temperature, but an attempt was made to search the microscopic pictures from the room temperature test for the irregular BD features. The data used in the search came from the high-voltage test with different inter electrode gaps (from 20 to 100 μm). Owing to the localization system, one can match the recorded data from the high voltage system to the specific event.

3. Results

3.1. High-voltage BD behavior during conditioning

A comparison of the conditioning curves for all the samples are shown in fig. 2 new picture and the main information for tests are in table 2.

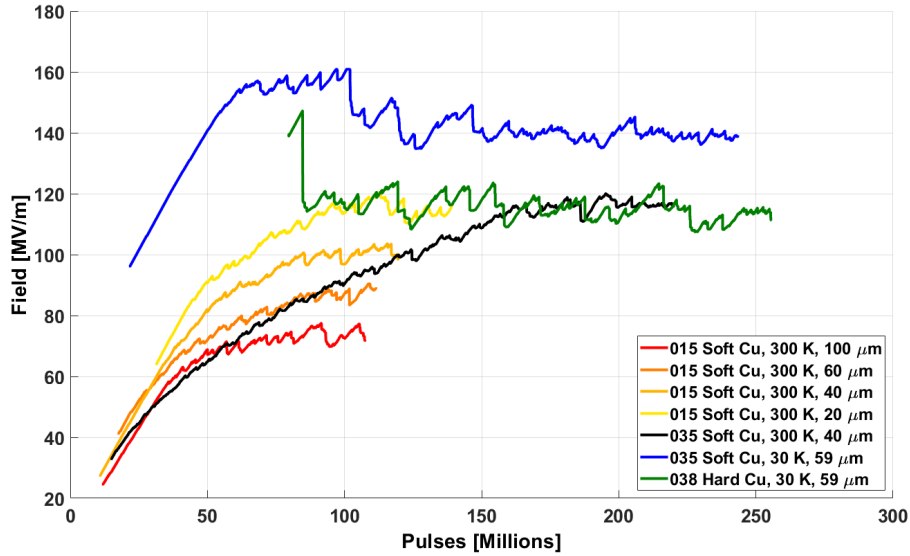


Figure 2: Comparison of the initial conditioning curves for 038 Hard Cu at 30 K and 035 Soft Cu at 300 K, together with the further conditioning of 035 Soft Cu at 30 K.

3.2. Characteristics of common breakdown features

During microscopic surface analysis of the samples, the breakdown features found on the surface of the electrodes were categorized. The general topography of the BDs ranged from extreme shallow to rough, with some intermediate morphology in-between. The anode spots always showed a shallow

Table 2: Main parameters for the tests.

Electrodes	Temperature [K]	Gap [μm]	Nb of pulses [$\times 10^6$]	Nb of Break-downs	Saturated electric field [MV/m]
015 Cu	300	100		516	
		60		931	
		40		681	
		20		236	
035 Cu	300	40	447	4121	120
	30	59	244	1963	158.8
038 Cu	30	59	255	1873	142.9
052 Cu	10	59	xxx	xxx	xxx

halo surrounding their rough circumference, with the central part of the spot always well-defined and reflective (fig. ??), which differed to the case of cathode spots (fig. ??).

Fig. 4a shows a typical crater left after a breakdown, with a halo around the center due to a capillary wave formed in a molten state. It is maximum 0.5 μm deep tiny grooves over all radius [7]. A typical round-shape anode BD feature (fig. 4a) is composed of *a reflective central part*, r_a (a green solid line), surrounded by *a wavy circumference*, w_a (a red dashed line), with *a thin lighter ring*, tlr_a (a purple square dotted line), around.

Meanwhile, a typical cathode BD feature has *a wavy dark major part*, w_c (a red dashed circle), and *a thin lighter ring*, tlr_c (a purple dotted circle), shown on fig. 4b. The cathode BD feature has usually 1 μm depth in the center (from reference [7]) and looks darker over all crater area (fig. 4), than on the anode. The machining grooves are not visible on BD cathode site, probably because of high temperature on site during BD and deep damage. We find emitted jets in random radial directions around BD crater, again typically only for cathode side (fig. 3b).

3.3. Star-like BD features

The investigation also discovered atypical topography between the BD sites, which were characterized by a very shallow depth with star-like shape instead of circular ones. The star-like BDs occur on the cathode only (fig. ??). No such BD was found on the anode. See fig. 5 for details of the star-like BD morphology.

More examples of atypical BD features are shown on fig. 6. The spots are spatially matched, with the top and bottom row showing features from the same breakdown on the anode respectively cathode. The same feature characterization was done for the atypical spots. Figs. 6a and 6d demonstrate how the regions for star-like BD features are defined, similarly to the regular BDs, described earlier (see figs. 4a and 4b).

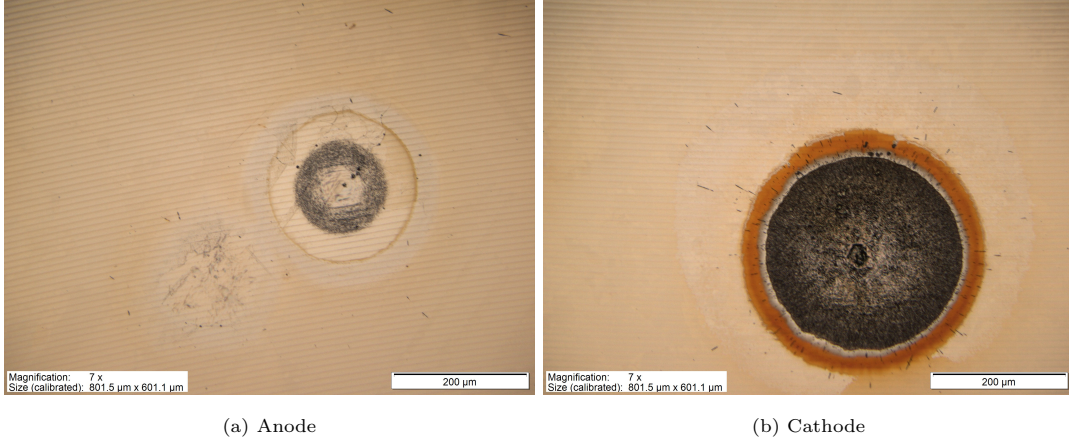


Figure 3: Typical BD features found on samples 035 and 038 : (a) a typical round-shape anode BD feature, (b) a typical cathode BD feature with visible jets around. Note: the images are not for the same BD event.

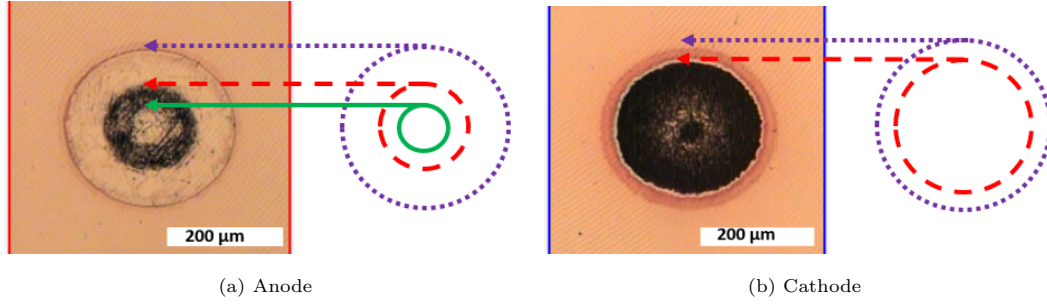


Figure 4: Typical BD features: (a) a typical round-shape anode BD feature is composed of a reflective central part (r_a , a green solid circle) surrounded by a wavy circumference (w_a , a red dashed circle) with a thin lighter ring (tlr_a , a purple square dotted circle) around; (b) a typical cathode BD feature has a wavy dark major part (w_c , a red dashed circle) and a thin lighter ring (tlr_c , a purple dotted circle).

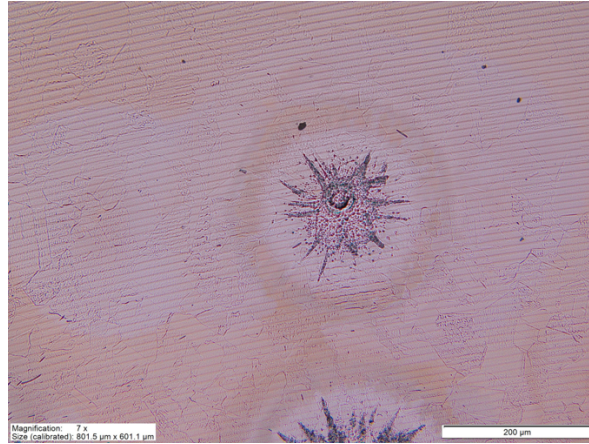


Figure 5: Details of the star-like BD morphology.

115 The anode BD feature corresponding to the same cathode spot looks very regular. One finds a
 116 typical light center with wavy halo around (fig. 6b). On average, the anode spots corresponding to
 117 star-like cathode sites are smaller and more shallow, with details hardly visible sometimes (fig. 6c).
 118 The machining grooves are also still visible under the feature.

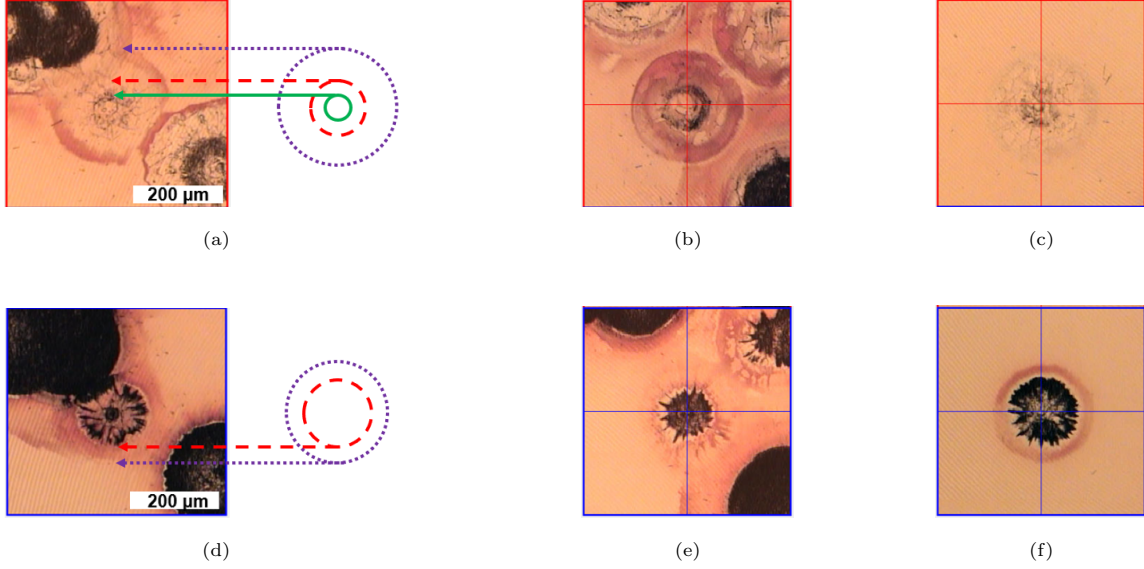


Figure 6: Breakdown features on anode (6a-6c) and corresponded atypical feature on cathode (6d - 6f) for hard Cu. (6a and 6d) represents similar to figs. 4a and 4b regions for star-like breakdown features: a typical round-shape anode BD feature is composed of a reflective central part (r_a , a green solid circle) surrounded by a wavy circumference (w_a , a red dashed circle) with a thin lighter ring (tlr_a , a purple square dotted circle) around; (d) a typical cathode BD feature has a wavy dark major part (w_c , a red dashed circle) and a thin lighter ring (tlr_c , a purple dotted circle). Figs. 6b-6c show more examples of anode features corresponded for 6e-6f star-like cathode feature.

119 3.4. Statistical analysis

120 3.4.1. Regular features

121 The ratios between different anode and cathode features were studies in details for the electrode
 122 038. The ratio between thin light rings radii of cathode and anode, tlr_c/tlr_a , was found to be 0.99 ± 0.08
 123 and the ratio between cathode and anode radii for wavy dark parts, w_c/w_a , to be 1.7 ± 0.54 . All the
 124 ratios between different anode and cathode features are summarized in table 3.

Table 3: Ratio between sizes of typical BD features for the 59? μm gap test with 038 Hard Cu.

No.	Ratio name	Number
1.	tlr_a/w_a	1.93 ± 0.21
2.	tlr_c/w_c	1.33 ± 0.11
3.	w_c/w_a	1.46 ± 0.17
4.	tlr_c/tlr_a	0.99 ± 0.08

Table 4: BD features on 038 and 052 electrodes (Ratio w_c/w_a)

Set	Feature type	Gap μm	Avg. radius, cathode w_c μm	Avg. radius, anode w_a μm	w_c/w_a
038	Regular	59	128 ± 2	80 ± 1	1.6 ± 0.1
052	Regular	59	114 ± 2	76 ± 1	1.4 ± 0.1
015	Regular	100	127 ± 2	82 ± 1	1.6 ± 0.2
015	Regular	60	101 ± 2	76 ± 1	1.4 ± 0.3
015	Regular	40	82 ± 2	70 ± 1	1.2 ± 0.3
015	Regular	20	61 ± 2	59 ± 1	1.1 ± 0.1

Sample 015 was studied at the pulsed dc system in CERN with the varied gap size. Afterwards, the morphological features were studied and the ratios w_c/w_a were measured for BD sites formed with different gap sizes (see table 4).

3.4.2. Crater size relation with E field and energy in the system

For sample 015 it was possible to correlate the breakdown voltage with the location of the breakdown spot, thus the relation between the size of the crater on anode and cathode could have been correlated with the electric field. See figure 7 for the comparison.

For the discussion:

The Cohesive Energy Rule [ANDRE ANDERS] can be formulated as follows: "The average arc burning voltage of a vacuum arc at a given current is approximately directly proportional to the cohesive energy of the cathode material" The arc burning voltage determines the energy dissipated for a given current, which is determined by the electrical discharge circuit.

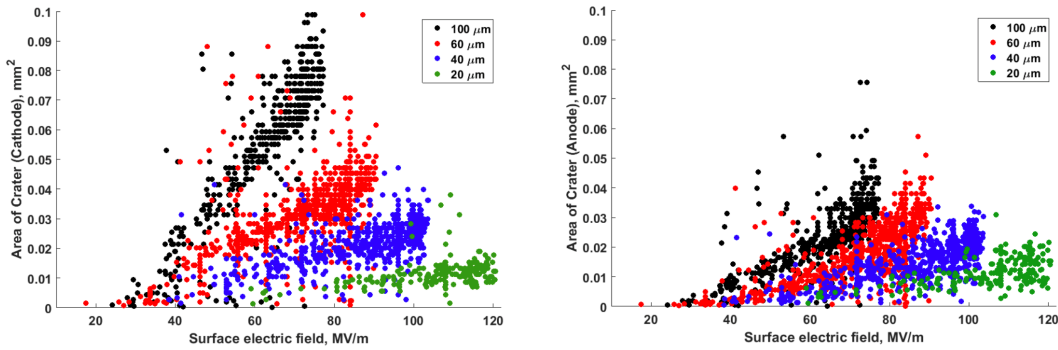


Figure 7: Comparison of breakdown crater sizes occurred with various surface electric field for different gap.

We can also calculate the energy in the system, assuming parallel plate capacitor model, consisting of two parallel conductive plates with area (A) separated with uniform gap (d) with a vacuum between ($\epsilon_0 = 8.85 \times 10^{-12}$ F/m). The gap d is much smaller than the electrodes dimensions. Our electrodes as a parallel plate capacitor can only store a finite amount of energy before breakdown occurs. The

141 maximum energy that they can store is¹:

$$E = \frac{1}{2} \epsilon_0 A d E^2 \quad (1)$$

142 In figure 8 the area of the crater calculated for anode and cathode is compared to the energy
143 calculated for the event using eq. 1.

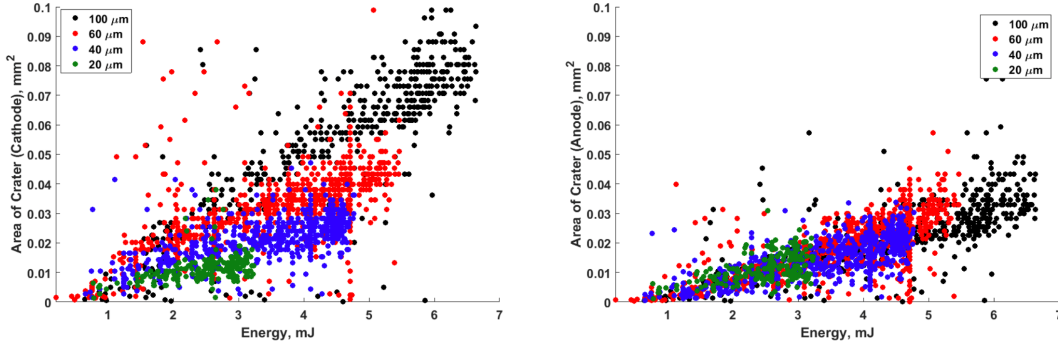


Figure 8: Breakdown energy dependence and its relation to the crater size.

144 3.4.3. Atypical features

145 The star-like features were predominately found on the samples that were tested at cryogenic
146 temperatures. Only a small percentage of such features were identified on sample 015 and all were
147 linked to experiments performed with the largest gap size, 100 μm. The BD features were counted and
148 linked to each other on both electrodes.

149 The comparison between number of counted BD couples of typical and atypical features are pre-
150 sented in table 5.

Table 5: Main statistical information about the microscopic images.

No.	Samples	Temperature	Number of BDs	Number of star-like couples
1.	015 Cu	300 K	516	8 (1.6 ± 1 %)
2.	035 Soft Cu	300 and 30 K	232	31 (13 ± 3 %)
3.	038 Hard Cu	30 K	145	37 (26 ± 5 %)
4.	052 Hard Cu	10 K	280	149 (53 ± 7 %)

151 After the initial analysis, a circle was fitted to each star-like spot to estimate the size. The com-
152 parison of the radii for anode and cathode spots is presented in table 6 and in fig 9. One can see that
153 the anode and cathode differ in radii, with the ration between 1.5-1.6. **For the discussion: The differ-**
154 **ence in the avg radius of star-likes for 038 and 052 sets could be explained with the aforementioned**
155 **dependence of the crater size and the energy. If the BD events in the 052 had access to more energy,**

¹We ignore, in this approximation, the energy change due to breakdown $\int_{\tau_{BD}}^{\tau_{BD}+\Delta\tau} I(t)V(t) dt$

we would expect bigger spots (on average). The ratio stayed the same. We should calculate the energy for the cryo sample 038 and 052 and, knowing area and gap size, recalculate average anode size from the linear relation obtained from set 015. Maybe we find something interesting there.

Table 6: Star-like BD features on 038 and 052 electrodes (Ratio w_c/w_a)

Set	Feature type	Gap μm	Avg. radius, cathode w_c μm	Avg. radius, anode w_a μm	w_c/w_a
038	Star-like	59	93 ± 3	58 ± 2	1.6 ± 0.1
052	Star-like	59	109 ± 2	76 ± 1	1.5 ± 0.1

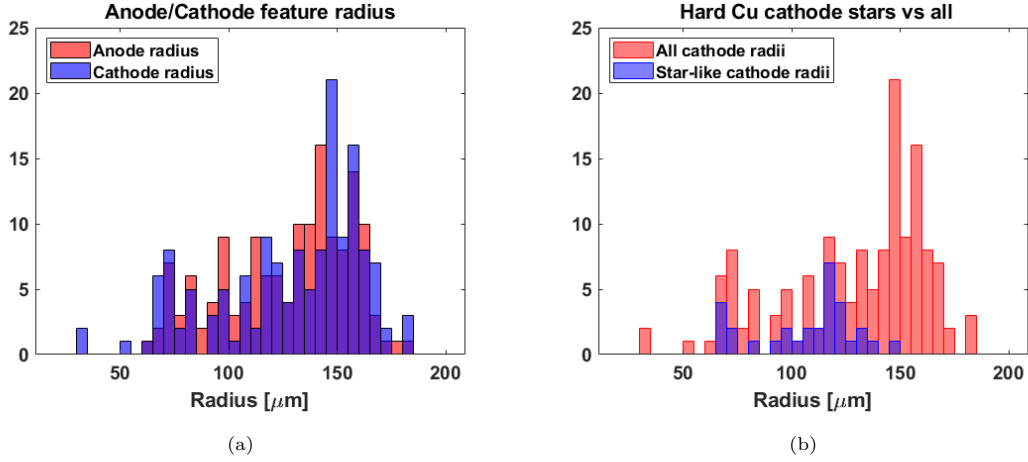


Figure 9: Comparison of radii for 038 Soft Cu: (a) anode and cathode for all BD features, (b) star-like and the rest of BD feature on the cathode.

Figure 9 includes results from the all features found on the surface where anode-cathode spots could be matched, including star-like spots. In fig. 9b the cathode radii between regular and irregular spots are compared. One can see that the spots dimension ranging between 70-150 μm and they are smaller, that for typical BD features.

The radii of BD features for 035 Soft Cu were in the same range as for 038 Hard Cu, between 20 and 180 μm , however the distribution is different. Most of the BDs have radii in range from 20 to 100 μm (fig. 10a), while for 038 Hard Cu is in a range 120 - 180 μm (fig. 9a). On fig. 10a, 2 peaks are visible, the second, small one, is in a 110 - 170 μm range. As it was mentioned before, with cold temperature the gap between electrodes increases leading to higher voltage BD voltage and also larger BD features on the surface². One can assume that the two different conditions of the test, at the room and at cold, resulted in two peak distribution. The first peak is related to warm conditions, and the second, smaller, is related to cold.

One can note that the radii from anode and cathode in the first peak do not match, the cathode

²The expanding plasma has larger volume to expand into, thus we observe larger radii for larger gap distances

seems to have smaller sizes. However, the atypical feature on the electrodes 035 Soft Cu have very similar size to those from 038 Hard Cu, i.e. between 50-150 μm .

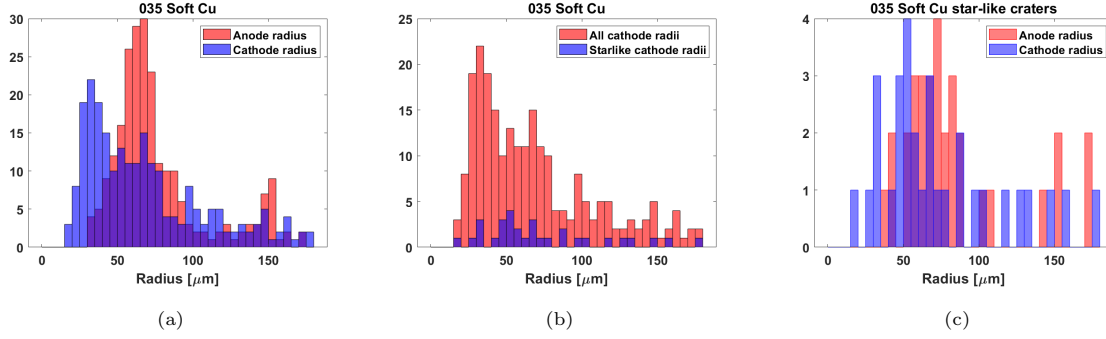


Figure 10: 035 Soft Cu.

3.5. Hardness

Vickers hardness test was performed on the different copper samples. The hardness for each sample was measured at several locations: the reference area with no high field activities during conditioning, in the area exposed to high field during conditioning, but without visible breakdown site and an area at the breakdown site. The last two measurements were done at the edge and at the center of the electrode. The results are presented in table 7. No significant difference between the hardness in the center of the electrode and on the edge was found. The hardness for soft and hard Cu samples was measured to 46-49 HV, and 107-110 HV respectively. We observed an increase of the hardness at the BD spot of the soft copper (30HV) and smaller decrease in hardness for the hard copper. The conditioning process doesn't change visibly the hardness outside of BD places.

Table 7: Hardness of copper, HV.

Material	Reference	Conditioned non-BD site		Conditioned BD site	
Place		center	edge	center	edge
Hard Cu	116.04 ± 3.64	107.74 ± 3.48	109.54 ± 3.27	97.29 ± 6.33	93.19 ± 3.26
Soft Cu	47.08 ± 1.0	48.57 ± 1.47	46.3 ± 1.10	76.46 ± 7.07	78.8 ± 6.9

3.6. Time-to-BD distribution and BD voltage study

In the dc system at CERN, equipped with a BD localization system, one can match the recorded data from the high voltage system to the specific event. With the recorded current, one can estimate "time-to-BD", defined as a time between a high voltage pulse start and a time when BD occurs in this pulse. The breakdown occurrence is easily identified as a short circuit detected by the high voltage generator. Fig. 11a shows the histogram of the time-to-BD for the data at different gap sizes for all the BDs. One can observe a rise of distribution around 0.8 μs for test with the 100 μm gap, while for

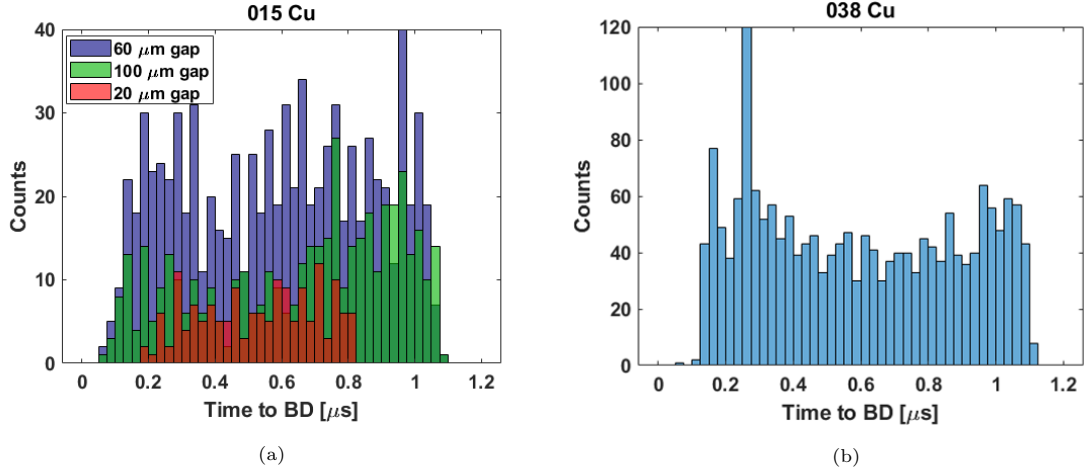


Figure 11: Time to BD distributions for the conditioning test: (a) in room temperature for 015 Cu with 100 (in green), 60 (in blue) and 20 (in red) μ m distance between electrodes. (b) for 038 Hard Cu at cryogenic temperature with 59 μ m between electrodes.

191 60 μ m gap is mostly uniform, and for 20 μ m is BDs occurred in the middle, not in the beginning or in
 192 the end of pulse.

193 In the end of the pulse, the high-voltage generator has already expended most of the energy and
 194 cannot sustain the discharge arc as long as for the breakdowns at the beginning of the pulse. One
 195 would therefore expect that if the star-like features are related to the available energy in the system,
 196 there could be a asymmetry in the time to BD distribution for the cold data. The distribution of time
 197 to BDs for the test in cold temperature (fig. 11b) also shows the rise of number of BDs in the end of
 198 the pulse, but it has also an excess of BDs in the beginning of pulse. Meanwhile, the distribution of
 199 time to BD for star-likes looks similar to the typical BDs (fig. 12).

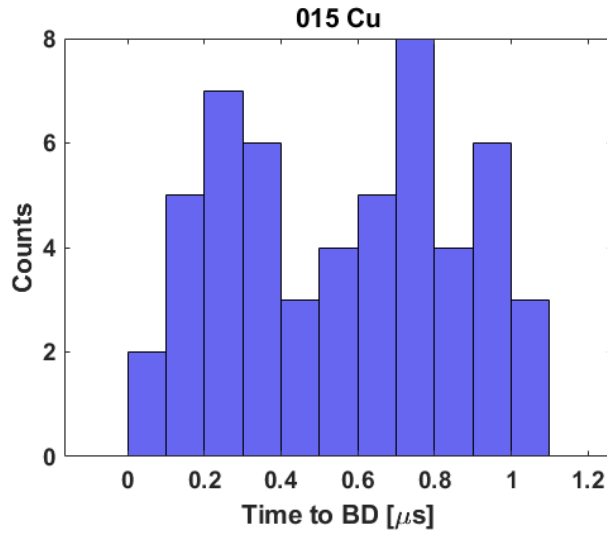


Figure 12: Time to BD for star-like BD feature occurred at room temperature test.

200 Another specific feature for the all star-like BDs in room temperature is the low BD voltage (fig. 13).

95% of star-like BDs have occurred during the ramping part of the conditioning, with 10-40% lower, than expected for this conditions, BD voltage.

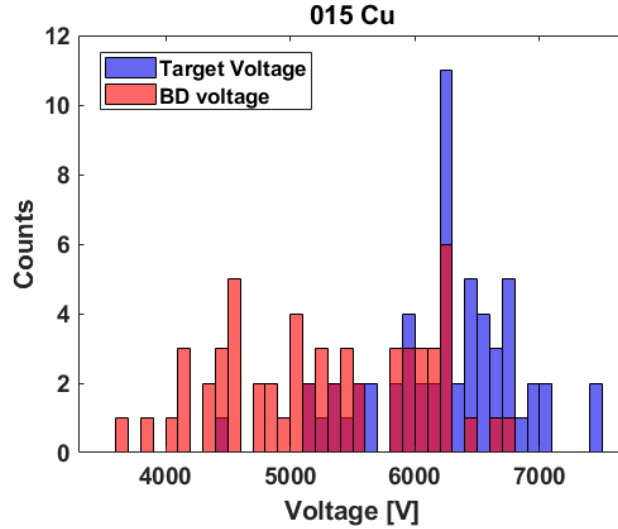


Figure 13: Target voltage vs Breakdown voltage for star-like features at for 015 Cu testes at room temperature.

4. Discussion

We should note more about the general aspect of the features. For example, the pristine surface of the copper has mechanical grooves that are visible. Which part of the crater cover(modify the groves? If we do not see the groves, that means the surface was melted. If we see - it was not melted, but it was covered by melted material. This if important as it will tell us something about temperature at this point.

H.Padamsee and J.Knobloch observed the starbursts on BD sites formed in superconducting RF cavity made from Nb and Nb electrodes tested under DC [8]. They proved with SEM and Auger spectroscopy that the starbursts were sites in which a very thin film of Fluorine contamination was burnt by plasma ignited during the BD event. Thus, the starburst was a fingerprint left by a plasma cloud in a thin contamination layer. The star-like BDs observed in the present work could also be the fingerprints of a plasma cloud. But they are deeper, i.e. they were left in the metal itself. Cu has much lower melting temperature than Nb: 1085 vs 2477°C and this could be the reason for deeper fingerprint. In the current study, the shallow plasma fingerprint left in the thin surface layer of contamination/oxidation was found beyond the rough circumference of the star-like BD.

Based on the SEM, FIB and TEM characterization data, the study concluded that the BD sites of atypical topography are most probably originate from tests done under cryo temperatures. During the conditioning of the electrodes, different temperature conditions were applied successively and there is no way to tell apart when a particular BD site was formed. In order to further investigate the star-like BD features, a dedicated high voltage test performed in one or two specific cryo temperature would be

223 very useful, to explained what specific conditions are needed to form such features and find out more
224 about their origin.

225 5. Conclusion

226 5.1. Possible physics behind the effect

227 Ideas from Andreas:

- 228 • The star burst is the ejection of molten material which (possibly due to various instabilities) is
229 emitted from jets in random directions, that lands on the surface and creating such shapes.
- 230 • The regular circle around the crater forms because of the material melting there, in an area much
231 bigger than the one covered by plasma (which is where the crater "pit" is), just because you need
232 some space to dissipate the ~ 1 kW (~ 20 V $\times$ ~ 50 A) of power concentrated in this small area.
233 If the ejected material from the eruption lands at this area, it will melt as well. Any kind of
234 shapes created by plasma instabilities will disappear, just because of Cu being liquid there. Thus,
235 in the room temperature experiments, there are no star-like traces left.
- 236 • In cryogenic temperatures, one would expect that this molten area is much smaller, because
237 of the significantly more efficient thermal dissipation. The thermal conductivity of Cu at, e.g.
238 30 K, is about 20 times higher than at room temperature. In this case, there is not always a
239 100 μ m-radius area of molten Cu around the crater, and bursts and droplets can land on solid
240 copper and swiftly crystallize on the surface, leaving clear traces.
- 241 • We can distinguish molten areas under microscope, since they are without grooves after machin-
242 ing.

243 5.2. Summary

- 244 • Difference in soft-copper and hard-copper: For sample 038 Hard Cu, $26 \pm \text{XX}\%$ of BD spots
245 were atypical, for sample 035 Soft Cu, assuming $\frac{1}{3}$ of breakdowns were made in cryo-conditions,
246 we get roughly 38%. Can difference in soft-copper and hard-copper explain it somehow?
- 247 • We can match anode-cathode sites to 100%, we only see star-like shape on cathode, matching
248 anode site is of regular, circular, morphology. This would make sense, since only the cathode is
249 expected to eject material.
- 250 • The size of the star-like feature are consistently smaller than regular one, on average 25%. The
251 corresponding anode sites are also on average smaller than "regular" sites.
- 252 • We see different distribution of BD spots radii for the electrodes conditioned at room and at
253 cold. The change in gap size could be one explanation.

• We observed star-like features on warm-conditioned electrodes at CERN, but only with 100 μm gap space and in a small fraction (2%) of cases. For smaller gaps (60 μm , 40 μm and 20 μm) no such features were observed.

We believe that it is worth to repeat the test at cold again, with bigger gap (60 μm instead of 40 μm). If the gap has a large effect, increasing probability to find a star-like feature, we should at least see increased statistics.

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